Practice workbook · Equation of a perpendicular line (including perpendicular bisectors)

Writing the equation of a line perpendicular to a given line (or segment) through a given point, including perpendicular bisectors and perpendiculars through a ratio point.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-COO-06

1. Question 10 · part b · 4 marks

2022 JAN · Paper 01

Worked-example retry · Attempt part b only.

10 The equation of the line L_1 is $y - 2x - 6 = 0$

The point P with coordinates $(2, a)$ lies on L_1

(a) Find the value of a

(1)

The line L_2 is perpendicular to L_1 and passes through P

(b) Show that an equation of L_2 is $x + 2y - 22 = 0$

(4)

Line L_1 crosses the x-axis at the point A and line L_2 crosses the x-axis at the point B

The point C has coordinates (m, n) such that $m > 0$ and $n < 0$

The length of AC is $5\sqrt{2}$ and the gradient of BC is $\frac{1}{4}$

(c) Find the value of m and the value of n

(9)

(d) Find the area of quadrilateral $ACBP$

(3)

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1. Question 10 · part b · 4 marks · Working and answer

Working and answer

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2. Question 2 · part a · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

2 The point A has coordinates $(-5, 3)$, the point B has coordinates $(4, 0)$ and the point C has coordinates $(-1, 5)$.

The line l passes through C and is perpendicular to AB.

(a) Find an equation of l.
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(4)

The line l intersects AB at the point D.

(b) Show that the coordinates of D are $(-2, 2)$.

(3)

(c) Show that l is not the perpendicular bisector of AB.

(2)

(d) Find the value of $\tan \angle ABC$.
Give your answer in its simplest form.

(4)

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2. Question 2 · part a · 4 marks · Working and answer

Working and answer

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3. Question 5 · part c · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

5 The line l with gradient $-\frac{1}{12}$ passes through the points A and B with coordinates $(p, 10)$ and $(123, 0)$ respectively.

(a) Show that $p = 3$

(2)

(b) Find an equation for l in the form $rx + sy + t = 0$ where r, s and t are integers.

(2)

The line k is perpendicular to l and passes through the point A.

(c) Find an equation for k in the form $y = mx + c$

(3)

Line k intersects the x-axis at the point C.

(d) Find the exact area of triangle ABC.

(4)

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3. Question 5 · part c · 3 marks · Working and answer

Working and answer

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4. Question 6 · part b · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part b only.

6 The line l passes through the point A with coordinates $(-2, 2)$ and the point B with coordinates $(3, 12)$
The point C with coordinates (p, q) lies on l such that $AC : CB = 3 : 2$

(a) Find the value of p and the value of q

(2)

The line k is perpendicular to l and passes through the point C

(b) Show that an equation of k is $2y + x - 17 = 0$

(4)

The line k crosses the x-axis at the point D

(c) Find the exact length of CD

(3)

The point X with coordinates (m, n) lies on l such that

$$\text{area of triangle } DXC = 80 \text{ units}^2$$

Given that $m > 0$

(d) find the value of m and the value of n

(7)

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4. Question 6 · part b · 4 marks · Working and answer

Working and answer

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5. Question 6 · part a · 5 marks

2020 JAN · Paper 01

Further practice · Attempt part a only.

6 The point A has coordinates (3, 0) and the point B has coordinates (2, 2).
The line L_1 passes through B and is perpendicular to AB.

(a) Find an equation of L_1

Give your answer in the form $ax + by + c = 0$

(5)

The line L_2 with equation $x - 7y - 3 = 0$ intersects the line L_1 at the point C.
The midpoint of AC is M.

(b) Find the coordinates of M.

(5)

(c) Find the area of the triangle ABM.

(4)

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5. Question 6 · part a · 5 marks · Working and answer

Working and answer

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6. Question 3 · part a · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

3 The point A has coordinates $(1, 7)$ and the point B has coordinates $(9, 3)$

The line l is the perpendicular bisector of AB

(a) Find an equation of l

(5)

The line l crosses the x-axis at the point C

(b) Find the area of the triangle ABC

(5)

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6. Question 3 · part a · 5 marks · Working and answer

Working and answer

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7. Question 2 · part a · 5 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

2 The point P has coordinates $(-7, 3)$ and the point Q has coordinates $(3, 18)$

The point R divides the line PQ in the ratio $2:3$

The line l passes through R and is perpendicular to the line PQ

(a) Find an equation of line l
Give your answer in the form $ay + bx + c = 0$ where a, b and c are integers.

(5)

The line l crosses the x-axis at the point S

(b) Find the area of triangle QRS

(5)

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7. Question 2 · part a · 5 marks · Working and answer

Working and answer

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8. Question 8 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

8 The points A and B have coordinates $(1, 5)$ and $(9, 9)$ respectively.

(a) Find an equation of line AB, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

The line l is perpendicular to AB and passes through the point X which lies on AB such that $AX : XB = 3:1$

(b) Show that an equation of l is $y = -2x + 22$

(5)

The point C has coordinates $(6, p)$

Given that C lies on l

(c) find the value of p

(1)

$ABCD$ is a parallelogram where the x coordinate of D is negative.

(d) Find the coordinates of the point D

(3)

(e) Find the area of the parallelogram $ABCD$

(4)

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8. Question 8 · part b · 5 marks · Working and answer

Working and answer

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9. Question 4 · part b · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

4 The point A with coordinates $(12, 14)$ and the point B with coordinates $(q, 2)$ where q is a constant, lie on the straight line with equation $3y - 2x - p = 0$ where p is a constant.

(a) Find the value of p and the value of q

(3)

The line L is perpendicular to AB and passes through the point X, which lies on AB such that $AX : XB = 1 : 2$

(b) Find an equation for L in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(6)

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9. Question 4 · part b · 6 marks

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Further practice · Attempt part b only.

Question 4 continued

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9. Question 4 · part b · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

Question 4 continued

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9. Question 4 · part b · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 9 marks)

9. Question 4 · part b · 6 marks · Working and answer

Working and answer

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10. Question 11 · part a · 6 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

11 An equation of the straight line l is $y - 3x = 3$

The point A on l lies on the y-axis.
The point B on l has coordinates $(10, b)$, where b is an integer.
The point C divides AB in the ratio $2:3$

The straight line k passes through C and is perpendicular to l

(a) Show that an equation of k is

$$3y + x - 49 = 0$$

(6)

The point D with coordinates (p, q), where q is positive, is such that AD is parallel to k and the length of AD is $12\sqrt{10}$

(b) Find the coordinates of D

(6)

The point E lies on k such that DE is parallel to the y-axis.
The point F lies on l such that DF is parallel to the y-axis.

(c) Find the exact area of triangle ECF.

(5)

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10. Question 11 · part a · 6 marks · Working and answer

Working and answer

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